SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

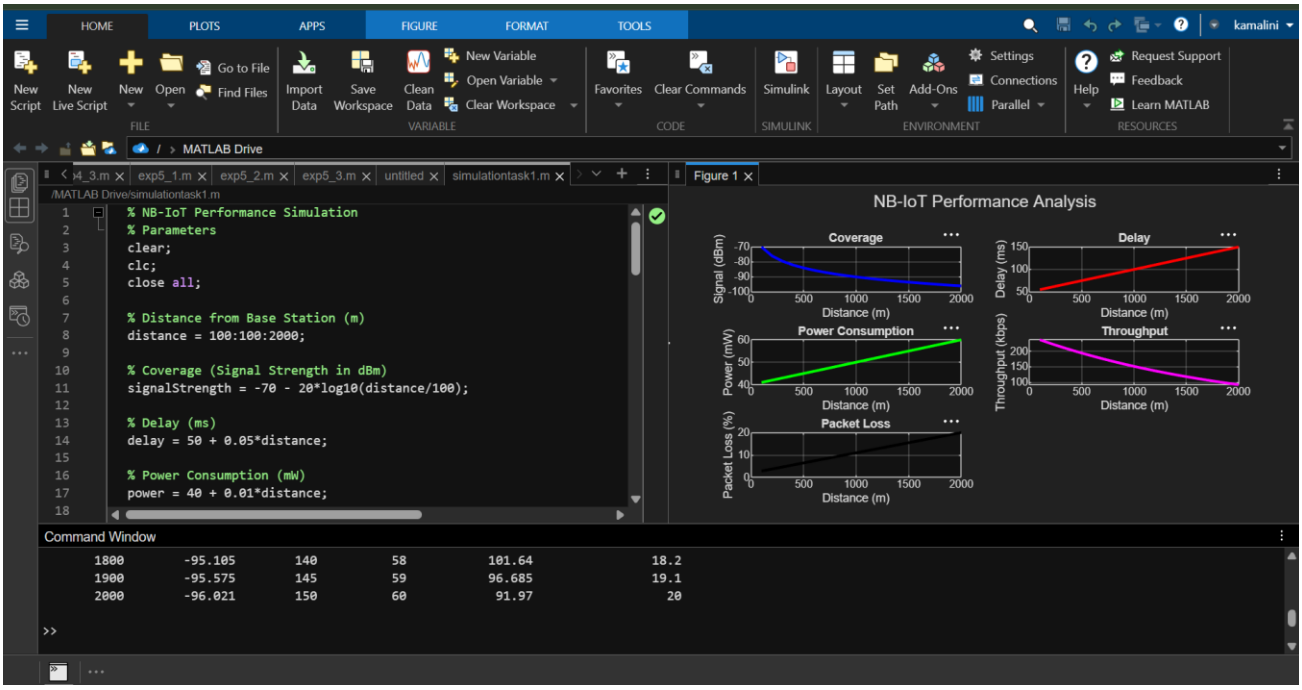
5G / 6G Communication Systems

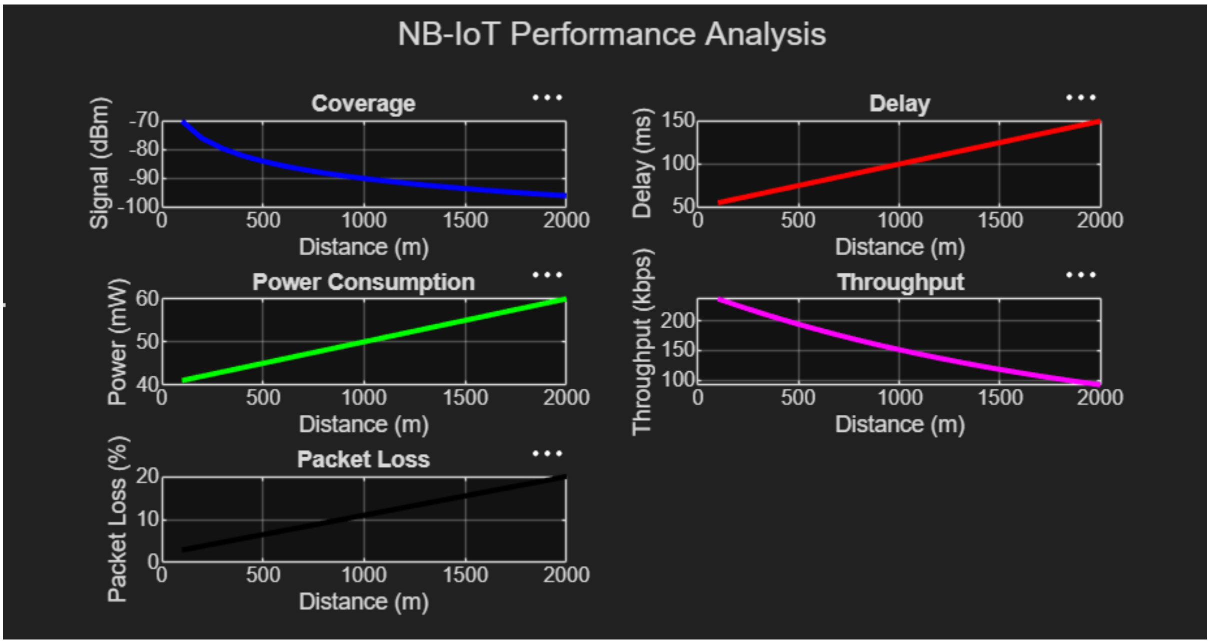
COURSE CODE: ECA56

1. Simulation of NB-IoT performance under coverage, delay, power consumption, throughput, and packet loss (BL4)

```
% NB-IoT Performance Simulation
% Parameters
clear;
clc;
close all;

% Distance from Base Station (m)
distance = 100:100:2000;

% Coverage (Signal Strength in dBm)
signalStrength = -70 - 20*log10(distance/100);

% Delay (ms)
delay = 50 + 0.05*distance;

% Power Consumption (mW)
power = 40 + 0.01*distance;

% Throughput (kbps)
throughput = 250*exp(-distance/2000);

% Packet Loss (%)
packetLoss = 2 + (distance/2000)*18;

% Display Results
T = table(distance', signalStrength', delay', power', throughput', packetLoss', ...
    'VariableNames', {'Distance_m','Signal_dBm','Delay_ms','Power_mW','Throughput_kbps','PacketLoss_percent'});

disp(T);

% Plot Results

figure;

subplot(3,2,1);
plot(distance, signalStrength,'b','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Signal (dBm)');
```

```
title('Coverage');

subplot(3,2,2);
plot(distance, delay,'r','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Delay (ms)');
title('Delay');

subplot(3,2,3);
plot(distance, power,'g','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Power (mW)');
title('Power Consumption');

subplot(3,2,4);
plot(distance, throughput,'m','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Throughput (kbps)');
title('Throughput');

subplot(3,2,5);
plot(distance, packetLoss,'k','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Packet Loss (%)');
title('Packet Loss');

sgtitle('NB-IoT Performance Analysis');
```



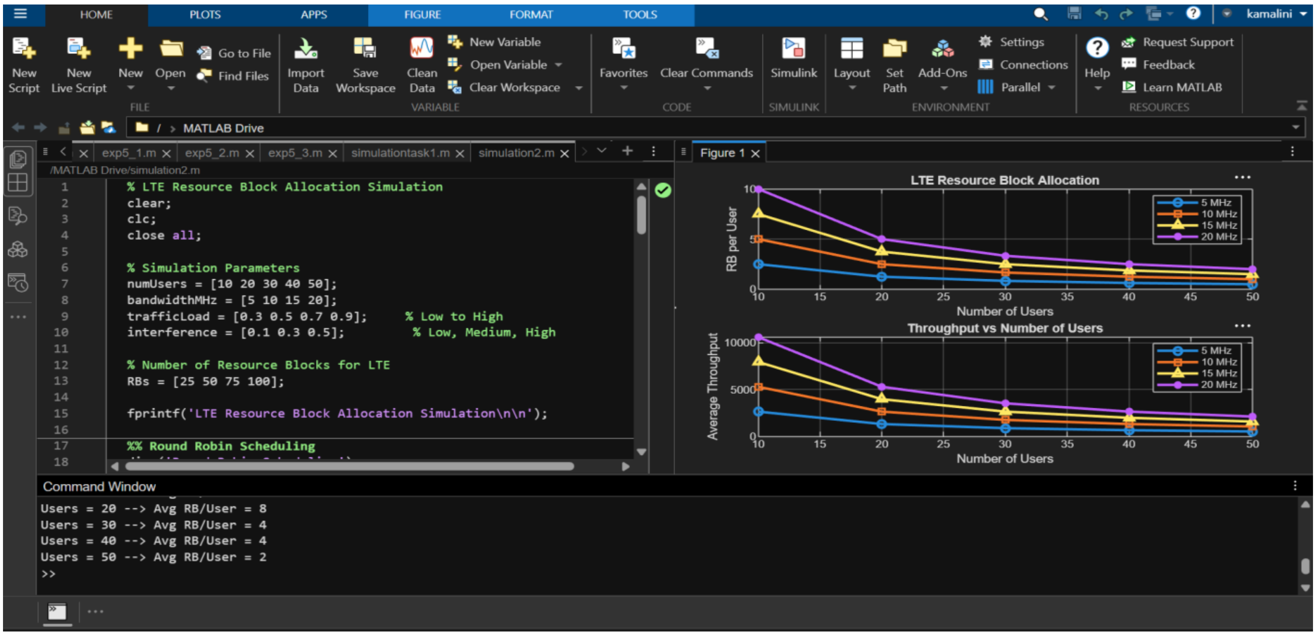


2. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels (BL4)

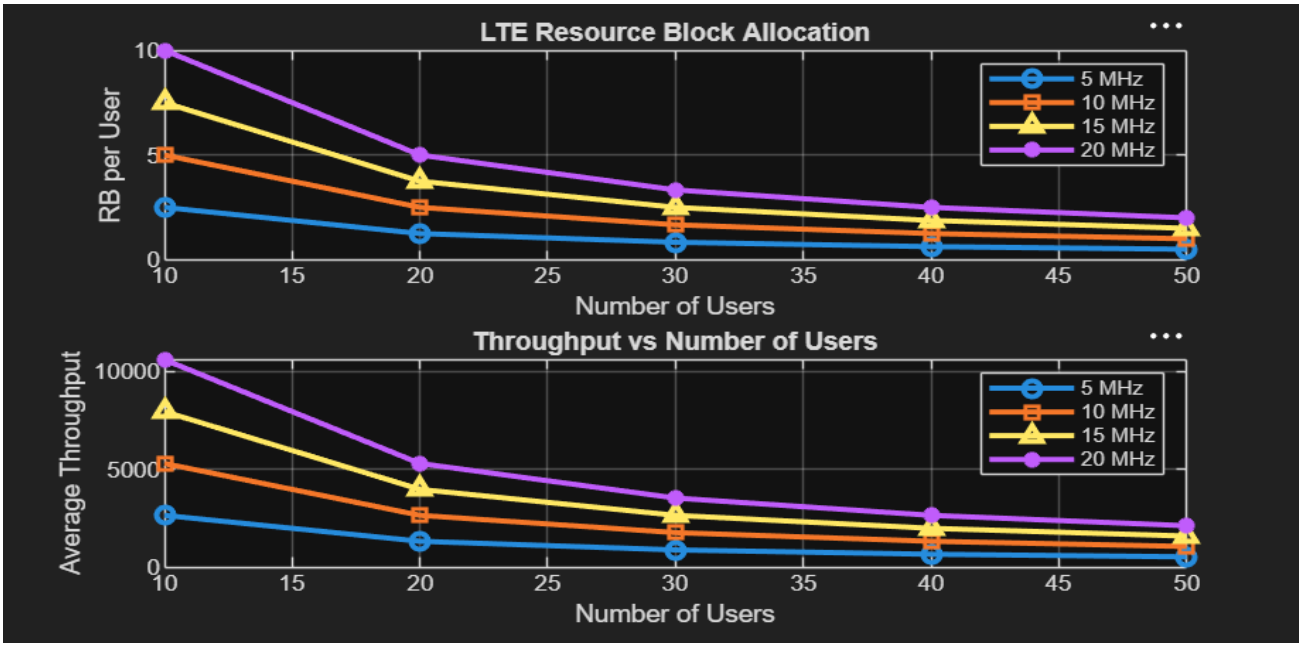
```
% LTE Resource Block Allocation Simulation

clear;

clc;

close all;

% Simulation Parameters

numUsers = [10 20 30 40 50];

bandwidthMHz = [5 10 15 20];

trafficLoad = [0.3 0.5 0.7 0.9];     % Low to High

interference = [0.1 0.3 0.5];       % Low, Medium, High

% Number of Resource Blocks for LTE

RBs = [25 50 75 100];

fprintf('LTE Resource Block Allocation Simulation\n\n');

%% Round Robin Scheduling

disp('Round Robin Scheduling');

for b = 1:length(RBs)

   fprintf('\nBandwidth = %d MHz\n', bandwidthMHz(b));

   for u = 1:length(numUsers)
```

```
        allocatedRB = floor(RBs(b)/numUsers(u));
        fprintf('Users = %d --> RB/User = %d\n', ...
            numUsers(u), allocatedRB);

    end
end

%% Proportional Fair Scheduling
disp(' ');
disp('Proportional Fair Scheduling');
for b = 1:length(RBs)
    fprintf('\nBandwidth = %d MHz\n', bandwidthMHz(b));
    for u = 1:length(numUsers)
        fairnessFactor = 1 + rand();
        allocatedRB = floor((RBs(b)/numUsers(u))*fairnessFactor);

        fprintf('Users = %d --> Avg RB/User = %d\n', ...
            numUsers(u), allocatedRB);

    end
end

%% Throughput Calculation
throughput = zeros(length(numUsers),length(bandwidthMHz));
for i = 1:length(numUsers)
    for j = 1:length(bandwidthMHz)
        throughput(i,j) = ...
            RBs(j)*180*12*(1-interference(2))*trafficLoad(3)/numUsers(i);
    end
end
%% Plot Resource Blocks
```

```
figure;
subplot(2,1,1)
plot(numUsers,RBs(1)./numUsers,'-o','LineWidth',2)
hold on
plot(numUsers,RBs(2)./numUsers,'-s','LineWidth',2)
plot(numUsers,RBs(3)./numUsers,'-^','LineWidth',2)
plot(numUsers,RBs(4)./numUsers,'-*','LineWidth',2)
grid on
xlabel('Number of Users')
ylabel('RB per User')
title('LTE Resource Block Allocation')
legend('5 MHz','10 MHz','15 MHz','20 MHz')
%% Plot Throughput
subplot(2,1,2)
plot(numUsers,throughput(:,1),'-o','LineWidth',2)
hold on
plot(numUsers,throughput(:,2),'-s','LineWidth',2)
plot(numUsers,throughput(:,3),'-^','LineWidth',2)
plot(numUsers,throughput(:,4),'-*','LineWidth',2)
grid on
xlabel('Number of Users')
ylabel('Average Throughput')
title('Throughput vs Number of Users')
legend('5 MHz','10 MHz','15 MHz','20 MHz')
```



HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
kamalini
MATLAB Drive
exp5_1.m
exp5_2.m
exp5_3.m
simulationtask1.m
simulation2.m
Figure 1
/MATLAB Drive/simulation2.m
% LTE Resource Block Allocation Simulation
clear;
clc;
close all;
% Simulation Parameters
numUsers = [10 20 30 40 50];
bandwidthMHz = [5 10 15 20];
trafficLoad = [0.3 0.5 0.7 0.9]; % Low to High
interference = [0.1 0.3 0.5]; % Low, Medium, High
% Number of Resource Blocks for LTE
RBs = [25 50 75 100];
fprintf('LTE Resource Block Allocation Simulation\n\n');
%% Round Robin Scheduling
Command Window
Users = 20 --> Avg RB/User = 8
Users = 30 --> Avg RB/User = 4
Users = 40 --> Avg RB/User = 4
Users = 50 --> Avg RB/User = 2
>>
LTE Resource Block Allocation
RB per User
Number of Users
Throughput vs Number of Users
Average Throughput
5 MHz
10 MHz
15 MHz
20 MHz



LTE Resource Block Allocation
RB per User
Number of Users
5 MHz
10 MHz
15 MHz
20 MHz
Throughput vs Number of Users
Average Throughput
Number of Users
5 MHz
10 MHz
15 MHz
20 MHz